INTERNATIONAL MONETARY FUND

ZIMBABWE

STAFF REPORT FOR THE 2014 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITY ANALYSIS

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The debt sustainability analysis (DSA) was prepared jointly by Bank and Fund staffs in accordance with the standardized Debt Sustainability Framework (DSF) methodology for Low-Income Countries (LICs).[1] *Zimbabwe remains in debt distress*

[1] Staff Guidance Note on the Application of the Joint Bank-Fund Debt Sustainability Framework for Low-Income Countries (http://www.imf.org/external/np/pp/eng/2013/110513.pdf).

BACKGROUND

1. Following the recent debt reconciliation exercise by the authorities for end-2011 and end-2012, Zimbabwe's total external debt has been estimated at about 70 percent of GDP (Text Tables 1 and 2). The Public and Publicly Guaranteed (PPG) external debt is broadly divided between bilateral creditors (60 percent) and multilateral creditors (40 percent). External private sector debt represents roughly 18 percent of total external debt. Compared to the 2012 DSA, there is a sharp downward revision of the total external debt stock ($2.5 billion), mainly driven by short-term private debt and supplier credits. The sharp revision to short-term private debt is caused by a change in assumptions: the BoP financing gap is no longer assumed to be covered with new short-term debt financing at the end of each year.

2. This change in assumptions for the financing gap follows a recent IMF technical assistance mission on multi-sector statistics. This mission recommended that any unidentified financing or financing gap left at the end of each year be consolidated in the errors and omissions line. Errors and omissions are driven by large under-recorded exports (mainly minerals, owing to efforts to avoid international restrictive measures), remittances through non-banking channels, and certain FDI-related imports (mining companies recapitalized in kind by parents who send in equipment but do not report the related financing).

3. Zimbabwe has improved its debt management capacity and the Debt Management Office (DMO) established at the Ministry of Finance and Economic Development (MoFED) is now fully staffed. The unit has developed a database for external debt that should facilitate debt management and promote information sharing between all government entities involved. However, decisions on new external borrowing continue to be taken with limited involvement of technical staff, with substantial risks of making sub-optimal choices with implications for medium-term sustainability.

Text Table 1. Zimbabwe: Total External Debt Stock by Creditor, 2011-12 [1/]

	2011		2012	
	In millions of USD	In percent of GDP	In millions of USD	In percent of GDP
Total	**8232**	**75.1**	**9031**	**72.4**
Public and Publicly Guaranteed Debt	**6268**	**57.2**	**6603**	**52.9**
Bilateral Creditors	3225	29.4	3476	27.9
Paris Club	2811	25.7	2881	23.1
Non-Paris Club	414	3.8	595	4.8
Multilateral institutions	2448	22.3	2525	20.2
IMF	135	1.2	127	1.0
AfDB	633	5.8	660	5.3
WB	1297	11.8	1348	10.8
EIB	305	2.8	313	2.5
Others	78	0.7	77	0.6
Short-term debt RBZ	595	5.4	603	4.8
Private Sector	**1964**	**17.9**	**2428**	**19.5**

Sources: WB, AfDB, Zimbabwean authorities, and staff estimates.
1/ For multilateral institutions, short-term debt, and suppliers' credits, estimates reflect a compound factor. Late interest is included under interest arrears.

Text Table 2. Zimbabwe: 2012 External Debt Stock by Servicing Status (in millions of US dollars) [1/]

	Remaining Principal Due	Total Arrears	Principal Arrears	Total Debt
Total	**3792**	**5240**	**3191**	**9031**
Public and Publicly Guaranteed Debt	1364	5240	3191	**6603**
Bilateral Creditors	838	2638	1432	**3476**
Paris Club	367	2513	1348	**2881**
Non-Paris Club	471	125	84	**595**
Multilateral institutions	525	1999	1157	**2525**
IMF	0	127	100	**127**
AfDB	55	606	306	**660**
WB	410	937	564	**1348**
EIB	34	279	152	**313**
Others	26	51	34	**77**
Short-term debt RBZ	0	603	603	**603**
Private Creditors	2428	0	0	**2428**

Sources: WB, AfDB, Zimbabwean authorities, and staff estimates.
1/ For multilateral institutions, short-term debt, and suppliers' credits, estimates reflect a compound factor. Late interest is included under interest arrears.

Box 1. Zimbabwe: Key Baseline Macroeconomic Assumptions

- The baseline scenario assumes average annual **real GDP growth** of about 4 percent for the projection period (2014–33), with somewhat lower growth at the beginning of the projection period (3 percent in 2014-15). In the medium term, uncertainty around the potential of diamond resources, constrained public finances, and the severe bottlenecks facing Zimbabwe would result in lower growth than in the recent past. Mining investment projects (especially in gold and platinum) should contribute to a change in the composition of growth and make up for sluggish growth in agriculture and manufacturing. The baseline assumes macroeconomic stabilization, but does not assume normalized access to finance or scaled-up public or private investment.

- **Inflation** is expected to be 2.5 percent over the medium term, accelerating from 0.2 percent in 2014.

- **Exports** are projected to grow at 7 percent over the medium term, in nominal terms.

- **Import** growth is projected to average 4 percent annually for the period 2014–19, and then increase to 5.5 percent until 2033.

- **The non-interest current account deficit is expected** to remain above 25 percent of GDP in 2014. Thereafter, it would decline to 15 percent in 2019 and stabilize to an average of 13 percent throughout the remaining projection period.

- **Tax revenues** are projected to remain broadly stable at around 27-28 percent of GDP. Gradual decline in customs revenues is expected to be offset by increases in other revenue sources.

- **External grants** are assumed to remain confined to humanitarian assistance, estimated in the range of 0.5 percent of GDP *per annum*. Under the baseline scenario, policies are unchanged and there is no debt relief.

- New **non-concessional borrowing** is assumed to average 1 percent of GDP annually during 2013-2016 and 2 percent of GDP from 2017 onwards,[1] with a grant element of about 27 percent. The assumption here is that Zimbabwe will continue to receive limited financing from non-traditional creditors in exchange for regular payments to them.

- While outstanding **external arrears** continue to be rolled over, the DSA assumes no accumulation of new external arrears.

1/ This assumption is based on loan disbursements, while the SMP ceiling monitors new non-concessional external borrowing contracted or guaranteed by the general government.

ASSUMPTIONS

4. The macroeconomic framework underpinning medium to long-term debt sustainability takes into account recent economic developments and progress in structural reforms pursued under the 2013-2014 Staff-Monitored Program (SMP). The medium-term projections are broadly consistent with the 2012 DSA, but some updates have been made to better reflect the consequences of the recent slowdown in emerging market economies (such as South Africa) and its impact on exports and FDI. The baseline scenario assumes no debt relief given limited progress in re-engaging with the main traditional creditors to date and the complexities inherent in making any other assumption. Therefore, in the baseline new external borrowing is expected to remain limited to resources provided by non-traditional creditors, mainly Brazil, China, and India (Box 1). Zimbabwe's CPIA rating has marginally improved since the 2012 DSA from 1.7 to 2.1.

5. Significant revisions have been made to the total external debt stock. The new estimates of external debt are based on the first results of the debt reconciliation exercise undertaken by the authorities. At end-2011, PPG external debt amounted to US$ 5.7 billion while private external debt reached US$ 1.7 billion after the change in the treatment of unidentified financing explained above. Compared to the 2012 DSA, total debt thus declined substantially, from 113 to 74 percent of GDP, largely due to this methodological change and to the recent substantial revision to the nominal GDP series.[2]

EXTERNAL DEBT SUSTAINABILITY

6. Even starting from a lower base than in the 2012 DSA, the PPG external debt overhang remains large (Table 1). Under the baseline, external debt is expected to continue to grow from 82 percent of GDP in 2013 to 122 percent in 2023, before declining to 108 percent in 2033. The present value (PV) of the PPG external debt will remain well above the 30 percent of GDP threshold during the entire projection period.[3] The PV of the PPG external debt-to-exports ratio will reach 171 percent in 2013 and decline to 119 percent 2023, and to 98 percent by the end of the projection period. The PV of the external debt-to-revenue ratio shows a downward trend, declining steadily from almost 176 percent in 2013 to about 138 percent in 2023 and to 114 in 2033. This is driven by the assumption of moderate debt accumulation on non-concessional terms (2 percent of GDP) mentioned above.

7. Debt service as a share of exports and revenues will remain below the respective thresholds throughout the entire projection period. The overall debt service-to-exports ratio will

[2] Zimbabwe's statistical agency (ZIMSTAT) recently revised the GDP series since 2009. The new GDP series are 25–30 percent higher than the old ones.

[3] The large residual is explained by the new assumption that any remaining financing gap in the balance of payments is treated as part of errors and omissions for both past and future years, rather than as generating new borrowing (as in the 2012 DSA).

remain close to 5 percent throughout the projection period, while the PPG debt service-to-exports ratio will remain around 2 percent until 2018, before slowly increasing to 3 percent in 2033.

8. These results remain broadly unchanged under alternative scenarios. The PV of the PPG external debt-to-GDP ratio is sensitive to lower GDP growth, reaching 69 percent in 2015 under the related bound test (Table 2a). In the case of a combined shock, the ratio will reach 88 percent in 2015, before declining to 44 percent in 2033. Under the lower exports scenario, the PV of the PPG external debt-to-exports ratio will reach almost 400 percent in 2015, before declining to just below 190 percent in 2033. Assuming lower FDI, the same ratio will reach 217 percent in 2015 and decline to around 102 percent by the end of the projection period. Similar results are obtained for all indicators under the lower GDP growth scenario and the less favorable borrowing terms scenario.

PUBLIC SECTOR DEBT SUSTAINABILITY

9. While public debt remains at high levels, the outlook is expected to improve (Table 3). The PPG debt-to-GDP ratio is projected to decline modestly from 62 percent in 2013 to 51 percent in 2023, and then to 41 percent in 2033. The PV of the PPG debt-to-revenue ratio shows a similar profile, declining from 206 percent in 2013 to 123 percent in 2033. The debt service-to-revenue ratio will remain around 3 percent until 2023 and then gradually increase to 5 percent by the end of the projection period.

Box 2. Domestic Public Debt in Zimbabwe

Overall domestic public debt in Zimbabwe is low and stable, at around 8-9 percent of GDP. However, this masks the diverging paths of its components. While the domestic debt of the Reserve Bank of Zimbabwe (RBZ) has been gradually shrinking, the stock of outstanding government securities has expanded, following the end of hyperinflation. Short-term domestic public debt has shrunk from 100 percent of the total in 2011 to 78 percent in 2013, as the government has managed to place securities with maturities of one year or more in the market. Finally, all of Zimbabwe's domestic public debt is denominated in U.S. dollars.

	Actual		
	2011	2012	2013
Total domestic public debt[1/]	887	1,110	1,117
Domestic debt of the RBZ	711	709	645
Government securities	0	188	314
Domestic payment arrears	176	213	158
Memorandum items:			
Total domestic public debt (percent of GDP)	8.1	8.9	8.6
Short-term domestic public debt (percent of total)[2/]	100.0	83.9	78.1

Sources: Zimbabwean authorities; and IMF staff estimates and projections.

1/ Millions of U.S. Dollars; unless otherwise indicated.
2/ Domestic public debt with original maturity of less than one year.

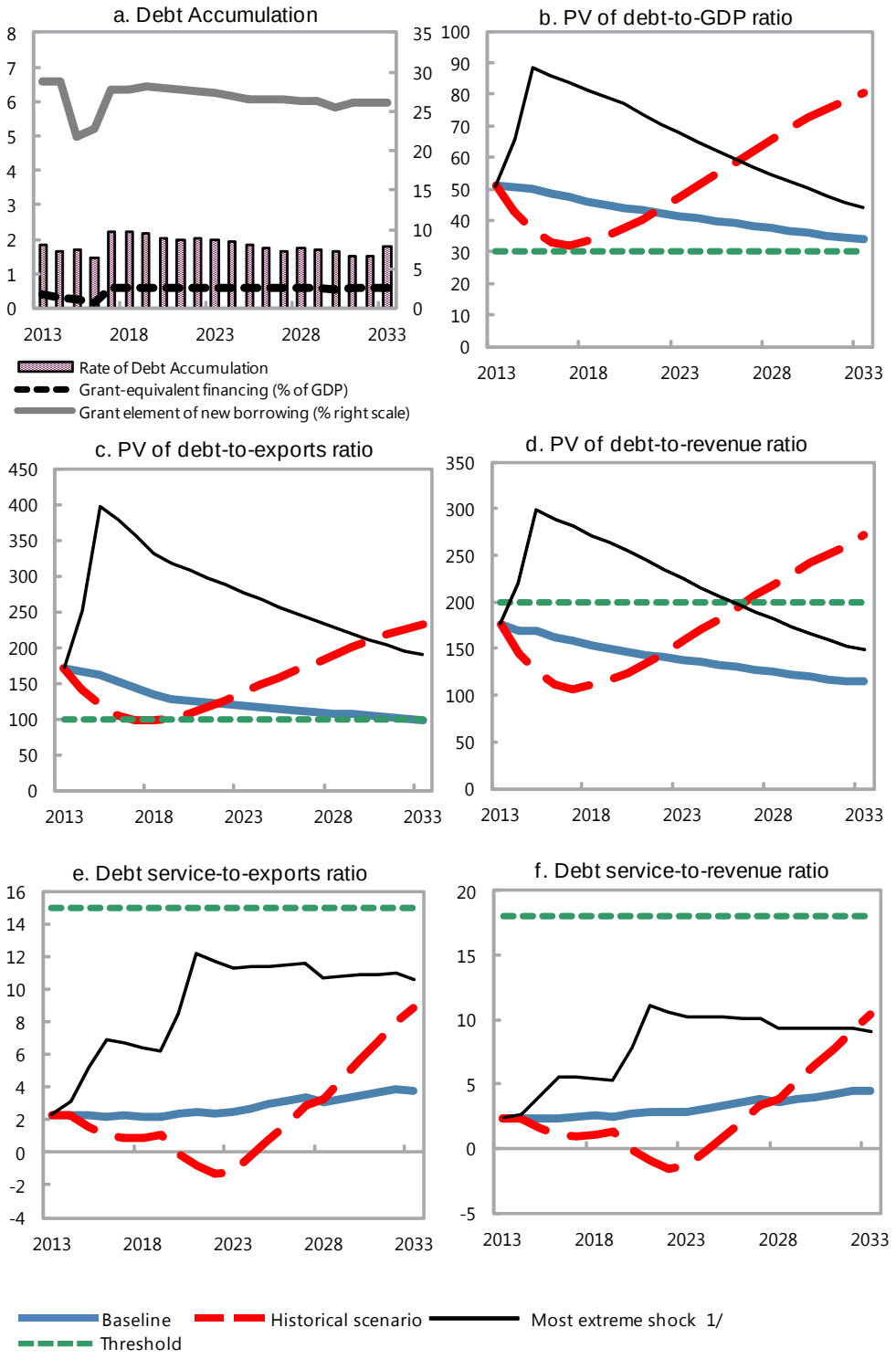
10. Alternative and shock scenarios show a similar gradually rising trend of debt indicators (Table 4)**.** Under the alternative scenario of lower GDP growth and primary balance at historical averages, the PV of the PPG debt-to-GDP ratio will increase from 59 percent in 2013 to 106 percent in 2023, and reach 175 percent by the end of the projection period. The PV of the PPG debt-to-revenue ratio will also deteriorate substantially under a low growth scenario. It goes from 206 percent in 2013 to above 500 percent by the end of the projection period. Finally, the debt service-to-revenue ratio will follow a similar path: a steady increase from 3 percent in 2013 to 32 percent in 2033. This reflects the fact that the scenario is based on historical averages for growth, which in Zimbabwe include a period of steep contraction in economic activity.

CONCLUSIONS

11. Zimbabwe remains in debt distress in the baseline scenario of no debt relief. In the baseline scenario, which includes new non-concessional borrowing of 2 percent of GDP a year on average from 2017 onwards, Zimbabwe's debt indicators remain above the relevant indicative thresholds. Zimbabwe's debt situation is vulnerable to shocks, particularly to exports and GDP growth. Zimbabwe's debt outlook also appears sensitive to FDI flows and to less favorable financial terms.

12. The Zimbabwean authorities share the staff assessment and therefore remain committed to re-engaging with all creditors, multilateral and bilateral. Staff encourages the authorities to continue to primarily seek grants and financing on terms as favorable as possible, ideally at highly concessional terms, to finance critical development projects with high economic returns. Limited non-concessional borrowing could be considered only if grants and concessional resources are unavailable or insufficient to implement critical growth-enhancing projects.

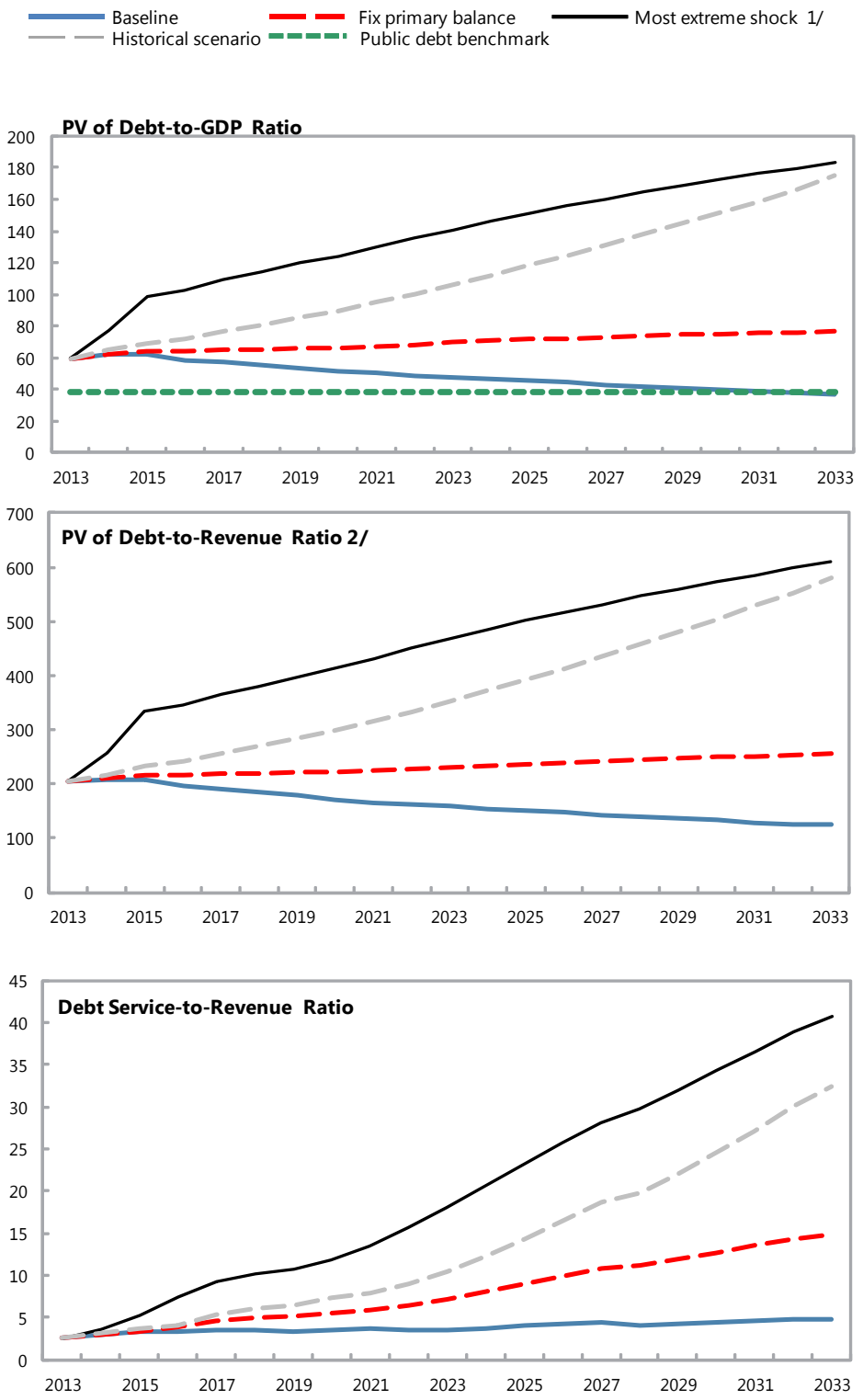
Figure 1. Zimbabwe: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2013-2033 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2023. In figure b. it corresponds to a Combination shock; in figure c. to an Exports shock; in figure d. to a Combination shock; in figure e. to an Exports shock; and in figure f. to a Combination shock.

Figure 2. Zimbabwe: Indicators of Public Debt Under Alternative Scenarios, 2013-2033 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2023.
2/ Revenues are defined inclusive of grants.

Table 1. Zimbabwe: External Debt Sustainability Framework, Baseline Scenario, 2010-2033 1/

(In percent, unless otherwise indicated)

	Actual			Historical 6/ Average	Standard 6/ Deviation	Projections									
	2010	2011	2012			2013	2014	2015	2016	2017	2018	2013-2018 Average	2023	2033	2019-2033 Average
External debt (nominal) 1/	**71.8**	**74.9**	**72.4**			**82.7**	**95.0**	**106.8**	**115.4**	**120.8**	**121.2**		**122.1**	**108.7**	
of which: public and publicly guaranteed (PPG)	60.6	57.0	53.6			53.4	52.8	52.4	50.6	50.0	48.9		44.9	37.8	
Change in external debt	-17.0	3.1	-2.5			10.3	12.3	11.8	8.6	5.4	0.4		-0.7	-1.3	
Identified net debt-creating flows	4.8	16.6	12.5			23.5	22.3	19.7	15.9	11.8	8.3		6.3	5.2	
Non-interest current account deficit	**17.6**	**29.5**	**24.3**	**16.3**	**13.1**	**28.5**	**27.4**	**26.3**	**23.3**	**19.9**	**16.3**		**15.0**	**12.3**	14.1
Deficit in balance of goods and services	24.8	35.6	30.2			33.3	31.9	30.8	28.4	25.4	22.4		18.9	13.9	
Exports	36.7	42.8	32.7			29.7	30.5	31.2	31.8	32.9	34.4		34.9	34.7	
Imports	61.5	78.4	62.9			63.0	62.4	62.0	60.1	58.3	56.8		53.7	48.6	
Net current transfers (negative = inflow)	-12.6	-14.0	-13.5	-8.3	5.0	-12.6	-11.5	-12.0	-11.8	-11.6	-11.1		-9.7	-7.6	-9.0
of which: official	-2.4	0.0	0.0			0.0	0.0	-0.4	-0.3	-0.3	-0.3		-0.3	-0.2	
Other current account flows (negative = net inflow)	5.5	8.0	7.5			7.8	7.1	7.4	6.8	6.2	5.1		5.7	6.0	
Net FDI (negative = inflow)	**-1.0**	**-3.3**	**-2.8**	**-1.2**	**1.1**	**-2.9**	**-2.6**	**-4.0**	**-3.9**	**-3.8**	**-3.7**		**-4.4**	**-3.4**	-4.0
Endogenous debt dynamics 2/	**-11.8**	**-9.6**	**-9.0**			**-2.1**	**-2.6**	**-2.6**	**-3.6**	**-4.3**	**-4.4**		**-4.3**	**-3.7**	
Contribution from nominal interest rate	0.3	0.3	0.1			0.2	-0.1	0.3	0.3	0.4	0.5		0.3	0.4	
Contribution from real GDP growth	-8.7	-7.4	-7.0			-2.3	-2.5	-2.9	-3.9	-4.7	-4.9		-4.6	-4.1	
Contribution from price and exchange rate changes	-3.4	-2.5	-2.2			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-21.8**	**-13.6**	**-15.0**			**-13.2**	**-9.9**	**-7.9**	**-7.3**	**-6.5**	**-7.9**		**-6.9**	**-6.5**	
of which: exceptional financing	2.5	-3.2	-1.7			-1.9	-0.3	-1.0	-1.0	-0.9	-0.9		-0.6	-0.7	
PV of external debt 4/	...	...	69.8			80.1	92.7	104.5	113.2	118.2	118.1		118.7	104.8	
In percent of exports	...	...	213.7			269.9	304.5	335.1	356.3	358.9	343.4		340.7	301.9	
PV of PPG external debt	**...**	**...**	**51.0**			**50.8**	**50.5**	**50.1**	**48.4**	**47.4**	**45.9**		**41.5**	**34.0**	
In percent of exports	**...**	**...**	**156.0**			**171.2**	**165.7**	**160.5**	**152.2**	**144.0**	**133.4**		**119.0**	**97.8**	
In percent of government revenues	**...**	**...**	**181.9**			**176.1**	**169.6**	**169.1**	**162.5**	**158.4**	**153.5**		**138.0**	**114.7**	
Debt service-to-exports ratio (in percent)	**6.5**	**3.8**	**3.3**			**7.9**	**6.2**	**5.2**	**4.9**	**4.5**	**3.9**		**4.1**	**4.0**	
PPG debt service-to-exports ratio (in percent)	**0.9**	**0.7**	**0.4**			**2.3**	**2.3**	**2.2**	**2.2**	**2.2**	**2.2**		**2.4**	**3.8**	
PPG debt service-to-revenue ratio (in percent)	**1.4**	**1.2**	**0.5**			**2.4**	**2.3**	**2.4**	**2.3**	**2.4**	**2.5**		**2.8**	**4.5**	
Total gross financing need (Billions of U.S. dollars)	2.3	3.7	3.6			4.7	5.1	5.6	6.2	6.8	7.1		10.7	19.6	
Non-interest current account deficit that stabilizes debt ratio	34.6	26.5	26.8			18.3	15.1	14.4	14.7	14.5	16.0		15.6	13.6	
Key macroeconomic assumptions															
Real GDP growth (in percent)	11.4	11.9	10.6	-1.2	11.1	3.3	3.1	3.2	3.9	4.3	4.4	3.7	4.0	4.0	4.0
GDP deflator in US dollar terms (change in percent)	4.0	3.7	3.0	5.2	8.5	0.7	0.8	1.1	2.5	2.3	3.6	1.8	2.5	2.5	2.5
Effective interest rate (percent) 5/	0.4	0.5	0.2	0.1	0.2	0.2	-0.1	0.4	0.3	0.3	0.5	0.3	0.3	0.4	0.3
Growth of exports of G&S (US dollar terms, in percent)	92.6	35.2	-13.2	10.4	31.8	-5.6	6.7	6.8	8.5	10.6	13.0	6.7	6.6	6.6	6.7
Growth of imports of G&S (US dollar terms, in percent)	-6.4	47.9	-8.7	17.3	35.7	4.1	2.9	3.7	3.2	3.4	5.4	3.8	4.9	4.5	5.6
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	28.9	28.9	21.9	22.9	27.8	27.9	26.4	27.3	26.1	26.8
Government revenues (excluding grants, in percent of GDP)	23.3	26.7	28.0			28.8	29.8	29.6	29.8	29.9	29.9		30.1	29.6	30.0
Aid flows (in Billions of US dollars) 7/	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
of which: Grants	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
of which: Concessional loans	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			0.4	0.3	0.2	0.2	0.6	0.6		0.6	0.6	0.6
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			28.9	28.9	21.9	22.9	27.8	27.9		27.3	26.1	26.8
Memorandum items:															
Nominal GDP (Billions of US dollars)	9.4	11.0	12.5			13.0	13.5	14.1	15.0	16.0	17.3		24.0	45.5	
Nominal dollar GDP growth	15.8	16.0	13.8			4.0	3.9	4.3	6.5	6.7	8.2	5.6	6.6	6.6	6.7
PV of PPG external debt (in Billions of US dollars)			6.4			6.6	6.8	7.0	7.2	7.6	7.9		9.9	15.5	
(PVt-PVt-1)/GDPt-1 (in percent)						1.9	1.7	1.7	1.5	2.2	2.2	1.9	2.0	1.8	1.8
Gross workers' remittances (Billions of US dollars)	0.4	0.6	0.6			0.8	0.8	0.8	0.9	0.9	0.9		1.2	2.0	
PV of PPG external debt (in percent of GDP + remittances)	...	...	48.5			48.0	47.8	47.2	45.7	44.9	43.5		39.5	32.5	
PV of PPG external debt (in percent of exports + remittances)	...	...	134.6			142.8	139.7	134.7	128.6	122.8	115.1		104.0	86.9	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	0.4			1.9	1.9	1.9	1.8	1.9	1.9		2.1	3.4	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2a. Zimbabwe: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013-2033

(In percent)

	Projections							
	2013	2014	2015	2016	2017	2018	**2023**	2033
PV of debt-to-GDP ratio								
Baseline	51	50	50	48	47	46	**41**	**34**
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	51	43	37	33	32	34	**48**	81
A2. New public sector loans on less favorable terms in 2013-2033 2	51	51	50	49	49	47	**45**	41
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	51	59	69	67	66	63	**57**	47
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	51	57	67	65	64	62	**52**	36
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	51	53	54	53	52	50	**45**	37
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	51	59	68	66	64	62	**52**	36
B5. Combination of B1-B4 using one-half standard deviation shocks	51	66	88	86	84	81	**68**	44
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	51	72	71	69	68	65	**59**	48
PV of debt-to-exports ratio								
Baseline	171	166	160	152	144	133	**119**	98
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	171	141	119	104	97	97	**136**	232
A2. New public sector loans on less favorable terms in 2013-2033 2	171	166	162	154	147	138	**129**	118
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	171	166	160	152	144	133	**119**	98
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	171	252	397	377	357	330	**276**	189
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	171	166	160	152	144	133	**119**	98
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	171	192	217	207	195	181	**151**	102
B5. Combination of B1-B4 using one-half standard deviation shocks	171	220	296	282	266	246	**203**	132
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	171	166	160	152	144	133	**119**	98
PV of debt-to-revenue ratio								
Baseline	176	170	169	163	158	153	**138**	115
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	176	145	126	111	107	112	**158**	272
A2. New public sector loans on less favorable terms in 2013-2033 2	176	170	170	164	162	158	**149**	138
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	176	199	234	225	219	212	**191**	159
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	176	190	227	219	213	206	**174**	120
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	176	177	184	177	172	167	**150**	125
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	176	197	229	221	215	208	**175**	120
B5. Combination of B1-B4 using one-half standard deviation shocks	176	220	299	288	281	272	**225**	149
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	176	241	241	231	226	219	**196**	163

Table 2b. Zimbabwe: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013-2033 (continued)

(In percent)

Debt service-to-exports ratio								
Baseline	2	2	2	2	2	2	**2**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	2	2	2	1	1	1	**-1**	9
A2. New public sector loans on less favorable terms in 2013-2033 2	2	2	2	2	2	2	**3**	6
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	2	2	2	2	2	2	**2**	4
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	2	3	5	7	7	6	**11**	11
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	2	2	2	2	2	2	**2**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	2	2	3	4	4	4	**6**	6
B5. Combination of B1-B4 using one-half standard deviation shocks	2	3	4	5	5	5	**9**	8
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	2	2	2	2	2	2	**2**	4
Debt service-to-revenue ratio								
Baseline	2	2	2	2	2	3	**3**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	2	2	2	1	1	1	**-1**	10
A2. New public sector loans on less favorable terms in 2013-2033 2	2	2	2	2	2	2	**4**	7
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	2	3	3	3	3	3	**4**	6
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	2	2	3	4	4	4	**7**	7
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	2	2	3	3	3	3	**3**	5
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	2	2	3	4	4	4	**7**	7
B5. Combination of B1-B4 using one-half standard deviation shocks	2	3	4	5	6	5	**10**	9
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	2	3	3	3	3	4	**4**	6
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	24	24	24	24	24	24	**24**	24

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is higher than in the baseline by 2 percentage points, while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios, except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3. Zimbabwe: Public Sector Debt Sustainability Framework, Baseline Scenario, 2010-2033

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate						Projections			
	2010	2011	2012	Average 5/	Standard Deviation 5/	2013	2014	2015	2016	2017	2018	2013-18 Average	2023	2033	2019-33 Average
Public sector debt 1/	67.5	65.1	62.5			62.0	64.4	64.1	60.9	60.0	58.1		51.0	41.1	
of which: foreign-currency denominated	67.5	65.1	62.5			62.0	64.4	64.1	60.9	60.0	58.1		51.0	41.1	
Change in public sector debt	-18.2	-2.5	-2.6			-0.4	2.4	-0.4	-3.1	-0.9	-1.9		-1.1	-0.6	
Identified debt-creating flows	-10.5	-6.9	-6.6			0.1	-0.2	-3.0	-4.7	-4.1	-5.1		-3.9	-3.3	
Primary deficit	0.8	2.1	1.1	1.9	1.4	2.3	2.1	-0.9	-1.4	-1.0	-1.4	-0.1	-1.2	-1.2	-1.2
Revenue and grants	23.3	26.7	28.0			28.8	29.8	29.6	29.8	29.9	29.9		30.1	30.1	
of which: grants	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Primary (noninterest) expenditure	24.1	28.7	29.2			31.1	31.8	28.7	28.3	29.0	28.5		28.9	28.8	
Automatic debt dynamics	-11.4	-9.0	-7.7			-2.2	-2.2	-2.0	-3.3	-3.1	-3.7		-2.7	-2.1	
Contribution from interest rate/growth differential	-9.3	-8.0	-7.1			-2.7	-2.7	-2.5	-2.9	-3.0	-2.9		-2.5	-1.9	
of which: contribution from average real interest rate	-0.6	-0.8	-0.8			-0.7	-0.8	-0.5	-0.5	-0.5	-0.4		-0.5	-0.3	
of which: contribution from real GDP growth	-8.8	-7.2	-6.2			-2.0	-1.9	-2.0	-2.4	-2.5	-2.5		-2.0	-1.6	
Contribution from real exchange rate depreciation	-2.0	-1.0	-0.7			0.5	0.4	0.5	-0.3	-0.1	-0.8		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-7.7	4.5	4.0			-0.5	2.6	2.6	1.6	3.2	3.2		2.9	2.7	
Other Sustainability Indicators															
PV of public sector debt	...	...	59.9			59.4	62.2	61.8	58.7	57.4	55.0		47.7	37.2	
of which: foreign-currency denominated	...	...	59.9			59.4	62.2	61.8	58.7	57.4	55.0		47.7	37.2	
of which: external	...	...	51.0			50.8	50.5	50.1	48.4	47.4	45.9		41.5	34.0	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	6.6	8.3	8.4			10.2	9.4	8.8	8.2	7.6	6.9		4.7	2.7	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	213.6			206.0	208.8	208.6	197.3	191.8	184.1		158.5	123.8	
PV of public sector debt-to-revenue ratio (in percent)	...	...	213.6			206.0	208.8	208.6	197.3	191.8	184.1		158.5	123.8	
of which: external 3/	...	...	181.9			176.1	169.6	169.1	162.5	158.4	153.5		138.0	113.0	
Debt service-to-revenue and grants ratio (in percent) 4/	1.4	1.2	0.6			2.5	3.0	3.4	3.4	3.5	3.5		3.5	4.8	
Debt service-to-revenue ratio (in percent) 4/	1.4	1.2	0.6			2.5	3.0	3.4	3.4	3.5	3.5		3.5	4.8	
Primary deficit that stabilizes the debt-to-GDP ratio	19.0	4.5	3.7			2.7	-0.3	-0.5	1.7	-0.1	0.5		-0.2	-0.6	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	11.4	11.9	10.6	-1.2	11.1	3.3	3.1	3.2	3.9	4.3	4.4	3.7	4.0	4.0	4.0
Average nominal interest rate on forex debt (in percent)	0.4	0.5	0.3	0.6	0.9	0.4	0.2	1.0	1.1	1.2	1.4	0.9	1.1	1.3	1.1
Real exchange rate depreciation (in percent, + indicates depreciation)	-2.6	-1.6	-1.2	-2.1	8.3	0.8	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	4.0	3.7	3.0	5.0	9.6	0.7	0.8	1.1	2.5	2.3	3.6	1.8	2.5	2.5	2.5
Growth of real primary spending (deflated by GDP deflator, in percent)	0.9	0.3	0.1	0.4	1.0	0.1	0.1	-0.1	0.0	0.1	0.0	0.0	0.0	0.0	0.0
Grant element of new external borrowing (in percent)	...	...	...	...	...	28.9	28.9	21.9	22.9	27.8	27.9	26.4	27.3	26.1	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 4. Zimbabwe: Sensitivity Analysis for Key Indicators of Public Debt, 2013-2033

	Projections							
	2013	2014	2015	2016	2017	2018	2023	2033
PV of Debt-to-GDP Ratio								
Baseline	59	62	62	59	57	55	48	37
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	59	65	69	72	77	81	106	175
A2. Primary balance is unchanged from 2013	59	62	64	64	65	65	69	77
A3. Permanently lower GDP growth 1/	59	64	66	66	69	70	92	190
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	59	77	99	103	109	114	141	183
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	59	63	66	63	61	59	51	39
B3. Combination of B1-B2 using one half standard deviation shocks	59	69	79	80	83	85	98	120
B4. One-time 30 percent real depreciation in 2014	59	87	85	81	78	74	62	46
B5. 10 percent of GDP increase in other debt-creating flows in 2014	59	70	69	66	65	62	54	41
PV of Debt-to-Revenue Ratio 2/								
Baseline	206	209	209	197	192	184	158	124
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	206	217	234	243	257	270	352	582
A2. Primary balance is unchanged from 2013	206	209	218	216	218	219	231	256
A3. Permanently lower GDP growth 1/	206	216	224	223	229	235	308	633
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	206	259	334	346	365	381	469	610
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	206	212	223	211	205	197	170	131
B3. Combination of B1-B2 using one half standard deviation shocks	206	232	266	269	278	285	327	400
B4. One-time 30 percent real depreciation in 2014	206	292	289	273	262	249	206	152
B5. 10 percent of GDP increase in other debt-creating flows in 2014	206	234	235	222	216	208	178	137
Debt Service-to-Revenue Ratio 2/								
Baseline	3	3	3	3	3	3	4	5
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	3	3	4	4	5	6	10	32
A2. Primary balance is unchanged from 2013	3	3	3	4	5	5	7	15
A3. Permanently lower GDP growth 1/	3	3	4	4	4	5	9	36
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	3	4	5	7	9	10	18	41
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	3	3	4	4	5	4	5	6
B3. Combination of B1-B2 using one half standard deviation shocks	3	3	4	5	6	7	11	25
B4. One-time 30 percent real depreciation in 2014	3	4	5	5	5	5	6	8
B5. 10 percent of GDP increase in other debt-creating flows in 2014	3	3	4	6	4	4	6	6

Sources: Country authorities; and staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

2/ Revenues are defined inclusive of grants.